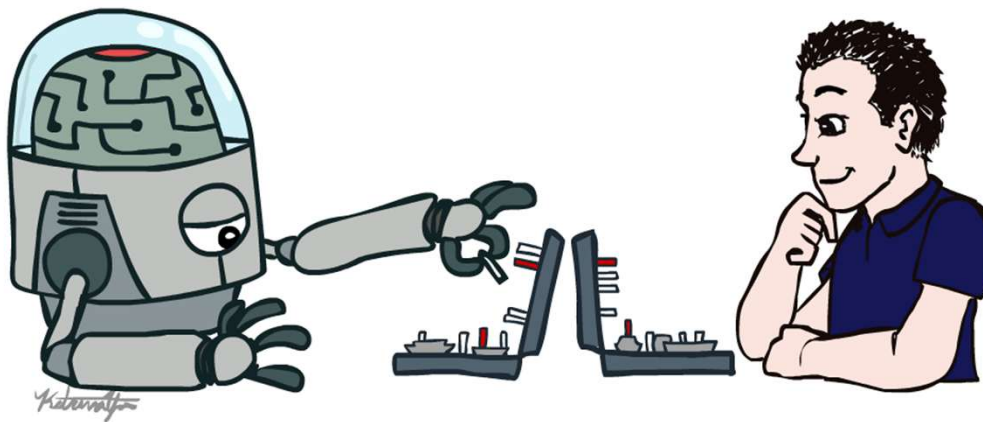


Introduction



KNTU, Spring 2025

[Slides are heavily borrowed from CS188 Intro to AI at UC Berkeley (ai.berkeley.edu).]

Resources

- Course website:

<https://sites.google.com/view/maryamabdolali/teaching/artificial-intelligence-spring-2025>

All resources (slides, codes, assignments, etc.) posted here

- TA email for course-related concerns: y.c.new.a@gmail.com

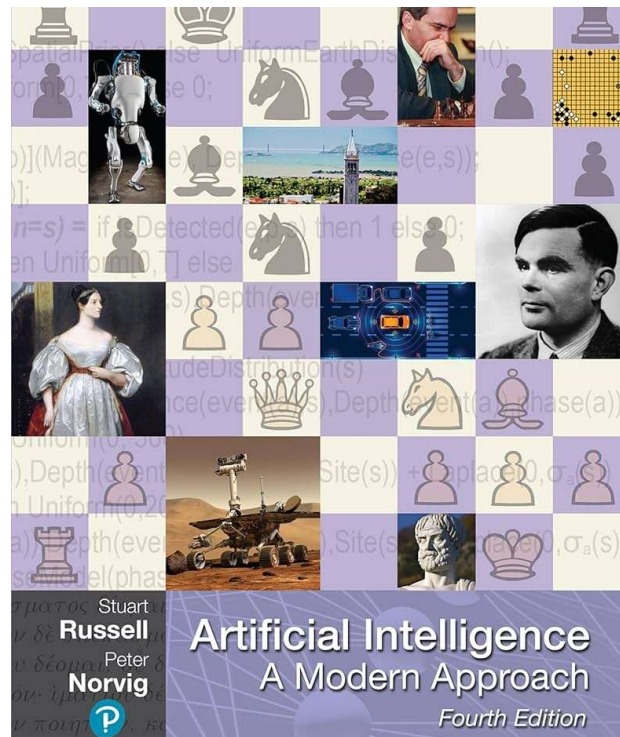
- <https://vc.kntu.ac.ir/>: Submit assignments here

- TAs:

- Majid Jafari
- Yasin Ahooyi
- Abbas Seifi

Textbook

- Not required, but for students who want to read more we recommend
 - Russell & Norvig, AI: A Modern Approach, 4th Ed.

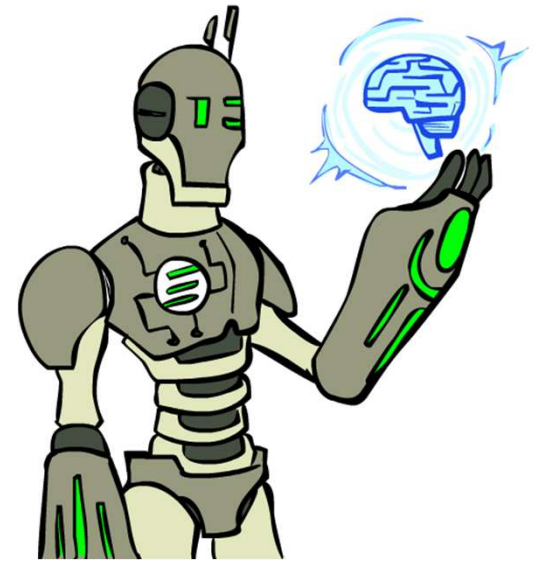


Grading Structure

- Final Project (25%)
- HomeWorks (25%)
- Final Exam (40%)
- Quiz (10%)
- Academic dishonesty policy
 - Negative points on assignments, and/or Fail in the class

What is AI?

- What is artificial intelligence?
- What can AI do?
- What is this course?



AI is having real-world impact

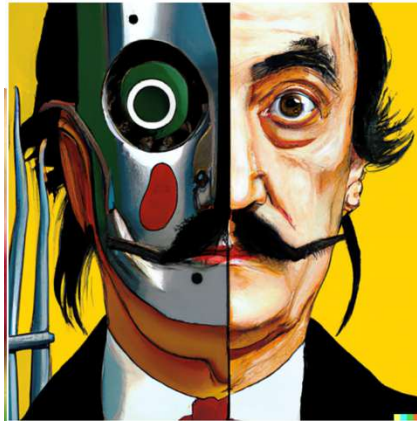
- Public imagination
 - Text assistants



AI is having real-world impact

- Public imagination

- Text assistants
- Image generation



vibrant portrait painting of Salvador Dalí with a robotic half face



a shiba inu wearing a beret and black turtleneck



a close up of a handpalm with leaves growing from it



an espresso machine that makes coffee from human souls, artstation



panda mad scientist mixing sparkling chemicals, artstation

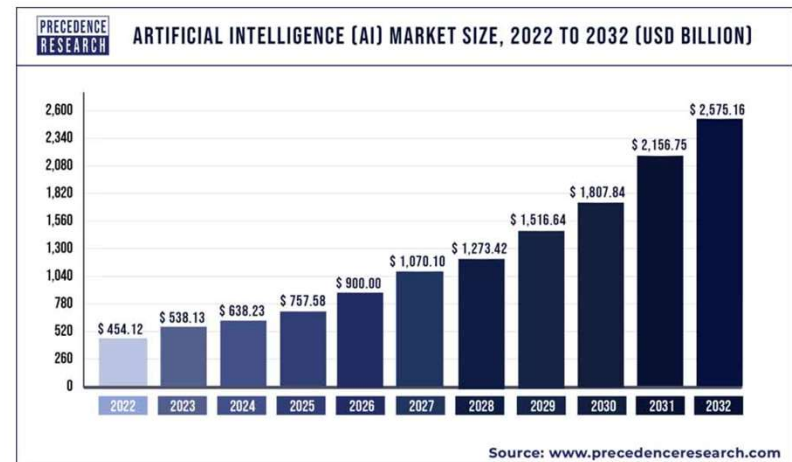


a corgi's head depicted as an explosion of a nebula

AI is having real-world impact

- Public imagination
- Economy
 - 454 billion USD globally

The global artificial intelligence (AI) market size was valued at USD 454.12 billion in 2022 and is expected to hit around USD 2,575.16 billion by 2032, progressing with a CAGR of 19% from 2023 to 2032. The North America artificial intelligence market was valued at USD 167.30 billion in 2022.



<https://www.precedenceresearch.com/artificial-intelligence-market>

AI is having real-world impact

- Public imagination
- Economy
- Politics



AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law

Aug. 18, 2023, 12:18 PM; Updated: Aug. 18, 2023, 12:48 PM

AI-Generated Art Lacks Copyright Protection, D.C. Court Says (1)



Riddhi Setty
Reporter



Isaiah Poritz
Legal Reporter



Bloomberg Law, 2023

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor

Finance & economics | Free exchange

New research shows the robots are coming for jobs—but stealthily

Look beneath the aggregate economic numbers, and change is afoot

The Economist, 2021

The Optimist's Guide to Artificial Intelligence and Work

The focus of much discussion is on how it will replace jobs, but nothing is inevitable.

New York Times, 2023

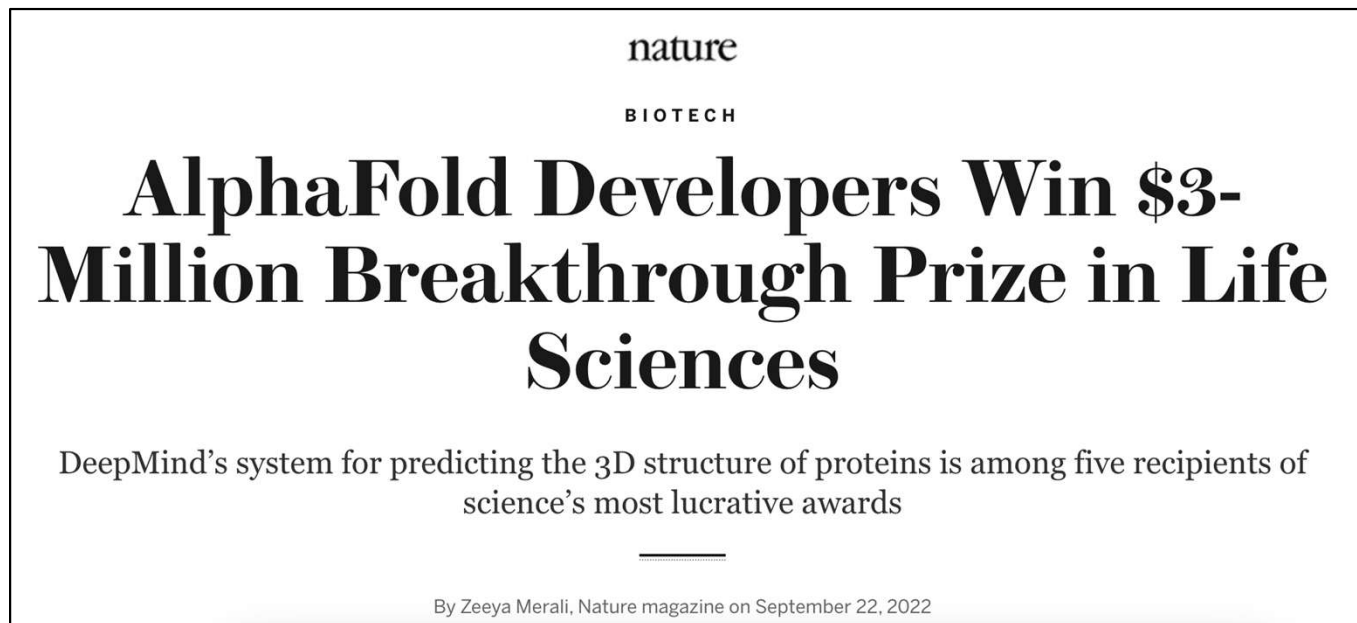
The human labor behind AI chatbots and other smart tools

Data labeling is an important step in developing artificial intelligence but also exposes the people doing the work to harmful content.

MarketWatch, 2023

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Sciences



Nature, 2022

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Sciences
- Education

BREAKING

ChatGPT In Schools: Here's Where It's Banned—And How It Could Potentially Help Students

Arianna Johnson Forbes Staff

I cover the latest trends in science, tech and healthcare.

Follow

2

Jan 18, 2023, 02:31pm EST

Forbes, 2023

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Sciences
- Education

Ok, but what actually is AI???

What is AI?

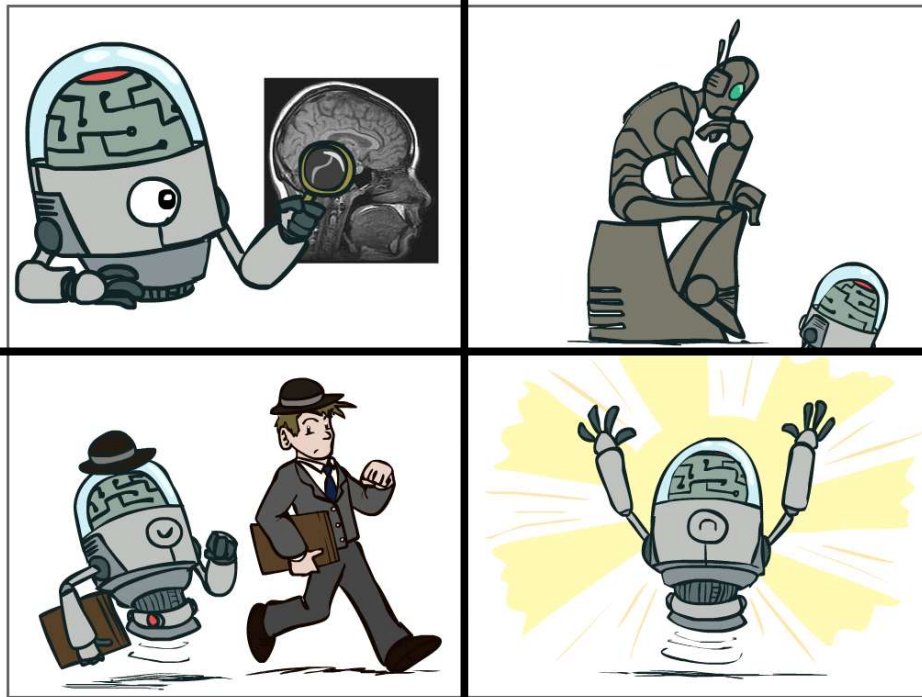
The science of making machines that:

we don't always
know how people
think

Think like people?

we have enough
humans! And there
is no need for a
machine to "have
favorite colors" like
humans

Act like people?



Have the right thought
process -> right
conclusions
You can think forever
but with no action,
useless!

Think rationally?

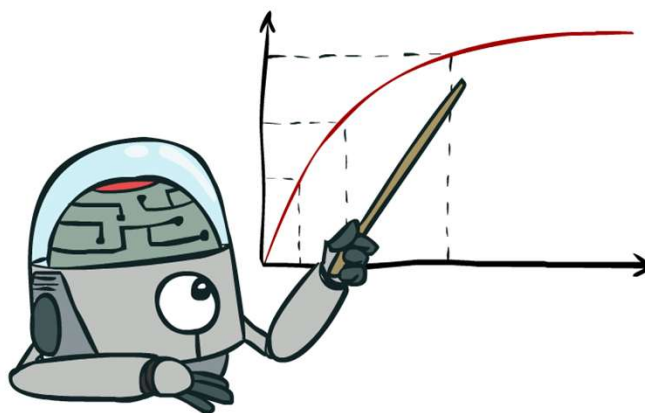
Key to all of it!
This Course!!

Act rationally?

Rational Decisions

- We'll use the term **rational** in a very specific, technical way:
 - Rational: *maximally achieving pre-defined goals*
 - Goals are expressed in terms of the **utility** of outcomes
 - World is uncertain, so we'll use **expected** utility
 - Being rational means **acting** to **maximize your expected utility**

Doesn't matter how you are thinking, as long as your actions lead to biggest expected utility

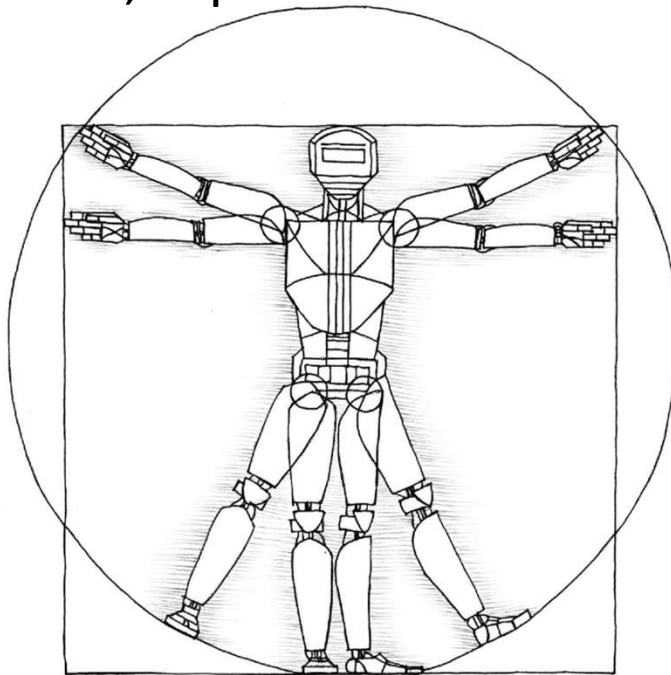


Perspectives on Intelligence

- Skills-based perspective
- “A system is only intelligent if it can do [X].”
 - Play chess?
 - Learn from experience?
 - Use words properly?
 - Make mistakes?
 - Not make mistakes?

Perspectives on Intelligence

- Embodiment perspective (Rodney Brooks)
 - contrasts this view with approaches that treat intelligence as a purely mental process, separate from the body and physical world.

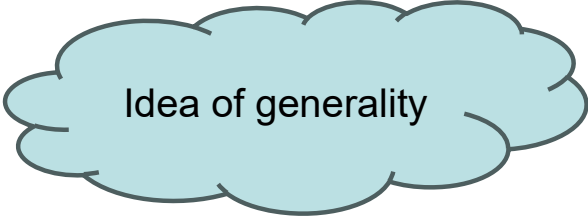


Perspectives on Intelligence

- Psychometrics perspective (François Chollet)
- “Measuring abilities, not skills [...] across a broad range of tasks, including tasks that were previously unknown to the ability-enabled system and its developers.”



Perspectives on Intelligence



Idea of generality

A human being should be able to change a diaper, plan an invasion, butcher a hog, conn a ship, design a building, write a sonnet, balance accounts, build a wall, set a bone, comfort the dying, take orders, give orders, cooperate, act alone, solve equations, analyze a new problem, pitch manure, program a computer, cook a tasty meal, fight efficiently, die gallantly. Specialization is for insects.

—Robert A. Heinlein

What About the Brain?

- Brains (human minds) are very good at making rational decisions, but not perfect
- Brains aren't as modular as software, so hard to reverse engineer!
- AI may be better than brains at some tasks
- *"Brains are to intelligence as wings are to flight"*
- Still, the brain can be a great inspiration for AI!

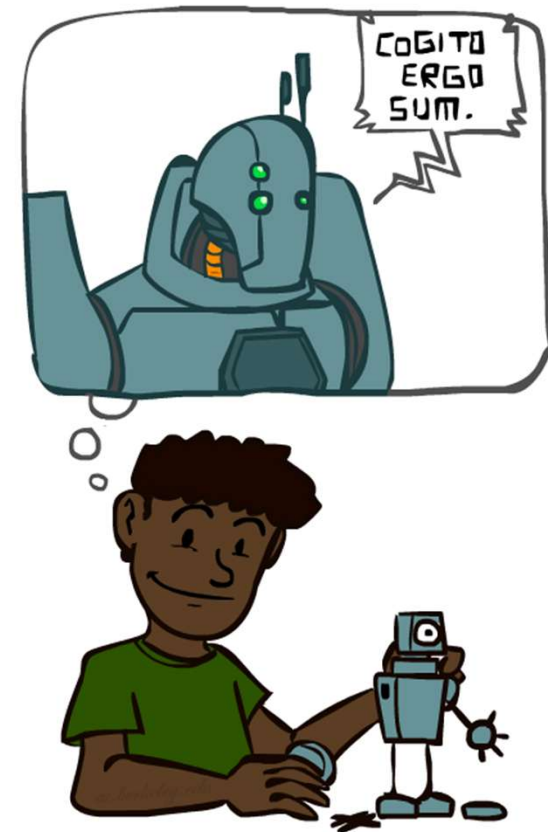


We sometimes do things
that we know we'd regret
later! :P

A (Short) History of AI

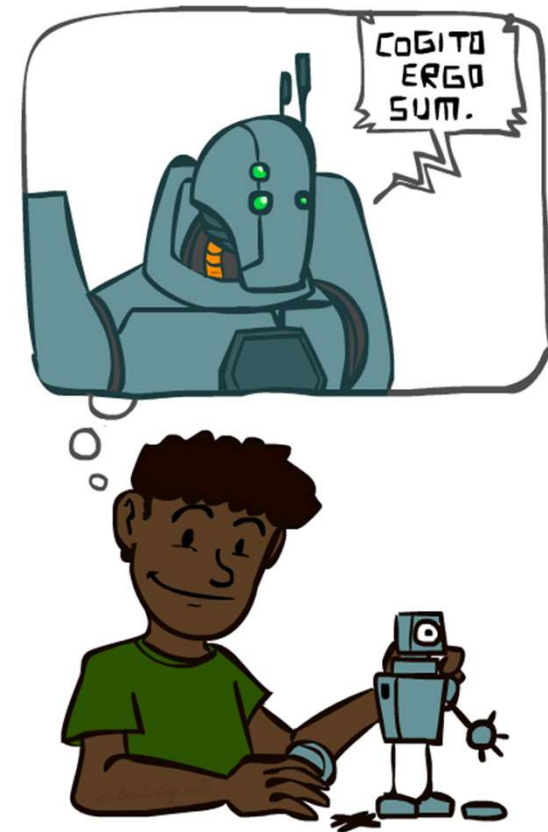
- **1940-1950: Early days: neural and computer science meet**
 - 1943: McCulloch & Pitts: Perceptron–boolean circuit model of brain: first artificial neuron
 - 1950: Turing's "Computing Machinery and Intelligence": Turing Test
- **1950—70: Excitement! Logic-driven**
 - 1950s: Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
 - 1956: Dartmouth meeting: "Artificial Intelligence" adopted

*"We propose that a **2-month, 10-man study of artificial intelligence** be carried out **during the summer of 1956** at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that **every aspect of learning** or **any other feature of intelligence** can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. **We think that a significant advance can be made** in one or more of these problems if a carefully selected group of scientists work on it together for a summer."*



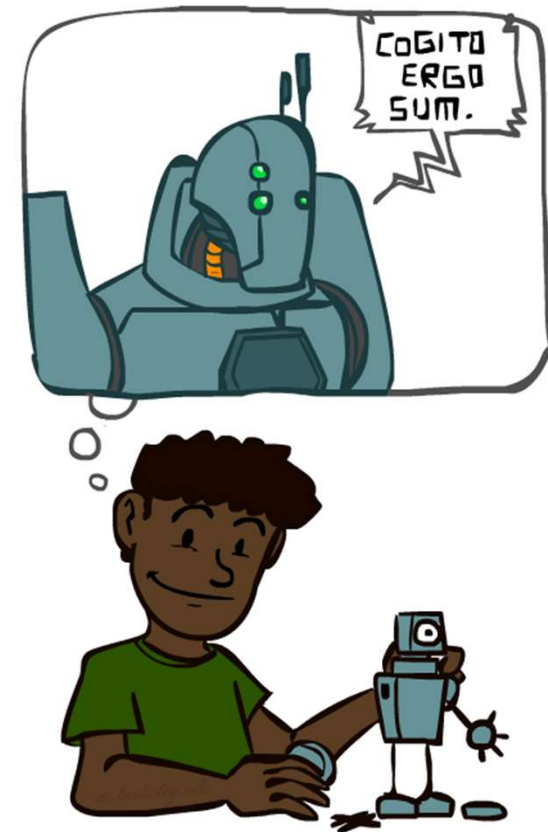
A (Short) History of AI

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 - 1943: McCulloch & Pitts: Perceptron—boolean circuit model of brain
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- **1950—70: Excitement! Logic-driven**
 - 1950s: Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
 - 1956: Dartmouth meeting: "Artificial Intelligence" adopted
 - 1969: Minsky & Papert: perceptrons can't learn XOR!
- **1970—90: Knowledge-based approaches**
 - 1969—79: Early development of knowledge-based systems
 - 1980—88: Expert systems industry booms; backpropagation makes it feasible to train multi-layer neural networks
 - 1988—93: Expert systems industry busts: "AI Winter"
- **1990—2010: Statistical approaches, agents**
 - Resurgence of probability, focus on uncertainty
 - Agents and learning systems... "AI Spring"?
 - 1992: TD-Gammon achieves human-level play at backgammon
 - 1997: Deep Blue defeats Gary Kasparov at chess
 - 2002: Embodied AI; Roomba vacuum invented



A (Short) History of AI

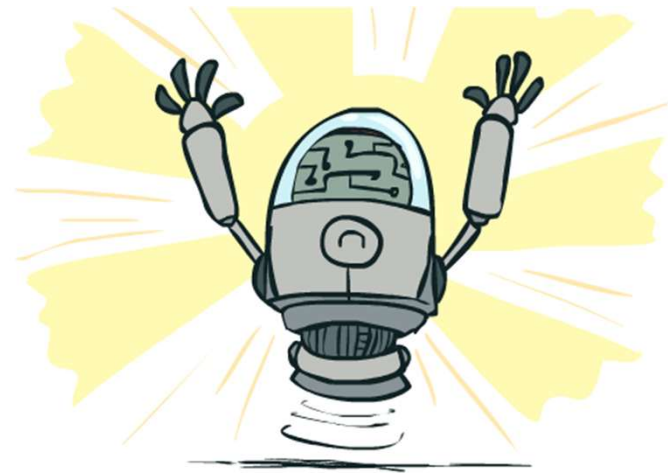
- 2010—2017: Big Data, GPUs, Deep Learning
 - 2011: Apple releases Siri
 - 2012: AlexNet wins ImageNet competition
 - 2015: DeepMind achieves “human-level” control in Atari games
 - 2016: DeepMind’s AlphaGo defeats Lee Sedol at Go
 - 2016: Google Translate migrates to neural networks
- 2017—: Scaling Up, Large Language Models
 - 2017: Google invents Transformer architecture
 - 2017: DeepStack defeat humans at poker
 - 2018-2020: AlphaFold predicts protein structure from amino acids
 - 2021-2022: Modern text-to-image generation
 - 2022: OpenAI releases ChatGPT
 - 2023: Every other company also releases a chatbot



What Can AI Do?

Quiz: Which of the following can be done at present?

- ✓ Win against any human at chess?
- ✓ Win against the best humans at Go?
- Play a decent game of table tennis?
- Unload any dishwasher in any home?
- Drive safely along the highway?
- Drive safely along streets of San Francisco?
- Buy a week's worth of groceries on the web?
- Discover and prove a new mathematical theorem?
- Perform a surgical operation?
- Translate spoken Chinese into spoken English in real time?
- Win an art competition?
- Write an intentionally funny story?
- Construct a building?



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Google's AlphaGo beats Lee Sedol at Go in 2016



But ... a plot twist in 2023!

DEEP BLUE WAS JUST THE START —

Man beats machine at Go in human victory over AI

Amateur exploited weakness in systems that have otherwise dominated grandmasters.

RICHARD WATERS, FINANCIAL TIMES - 2/19/2023, 4:51 AM

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2022 Google robot could make 340 consecutive hits.
But most hits were pretty easy...

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Discover and prove a new mathematical theorem?

Perform a surgical operation?

Translate spoken Chinese into spoken English in real time?

Win an art competition?

Write an intentionally funny story?

Construct a building?

California just opened the floodgates for self-driving cars

After a state regulatory board ruling, San Francisco will have 24/7 robotaxis on its streets. A Washington Post analysis shows how it's a pivotal moment for the industry.

By [Trisha Thadani](#) and [Jeremy B. Merrill](#)

Updated August 10, 2023 at 10:06 p.m. EDT | Published August 10, 2023 at 9:15 p.m. EDT

The Register

California DMV hits brakes on Cruise's SF driverless fleet after series of fender benders

50% chop effective 'immediately' as department investigates traffic and safety issues

[Jude Karabus](#)

Mon 21 Aug 2023 14:28 UTC

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- ? Write an intentionally funny story?
- Construct a building?



You

Please write me a very funny, and extremely short story.

< 6 / 6 >



ChatGPT

Once there was a chicken who was terribly afraid of crossing roads. One day, it mustered up all its courage, stepped onto the road, and halfway across, it suddenly stopped and said, "Wait, why am I doing this again?"

< 12 / 12 >    

What Can AI Do?

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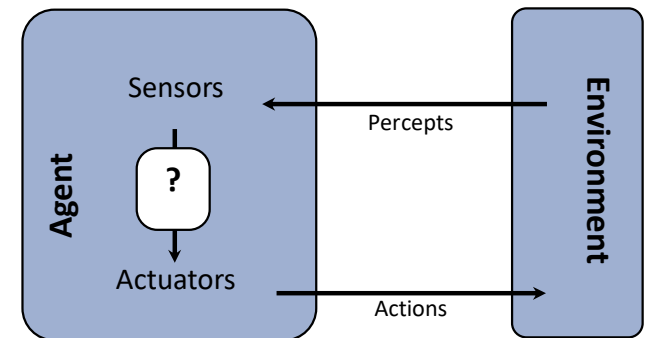
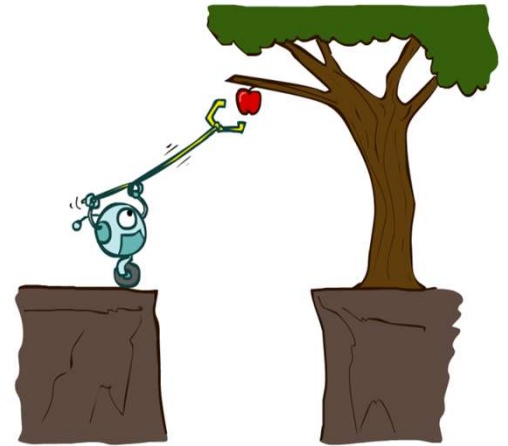
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- ✗ Construct a building?



This Course: Designing Rational Agents

- An **agent** is an entity that perceives and acts.
- A **rational agent** selects actions that maximize its (expected) **utility**.
- Characteristics of the **percepts**, **environment**, and **action space** dictate techniques for selecting rational actions



Course Topics

Core Components of Rational Agents:

Search &
Planning

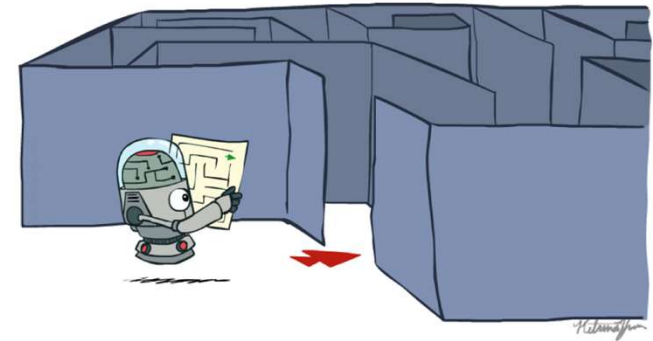
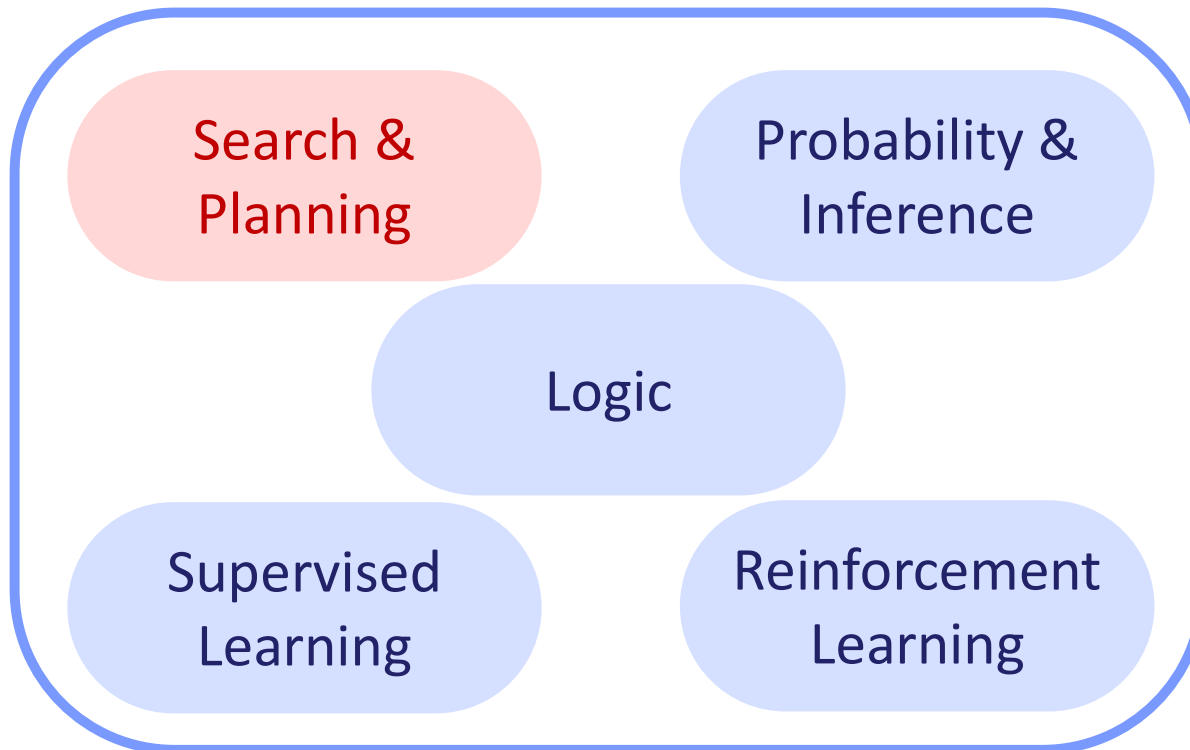
Probability &
Inference

Logic

Supervised
Learning

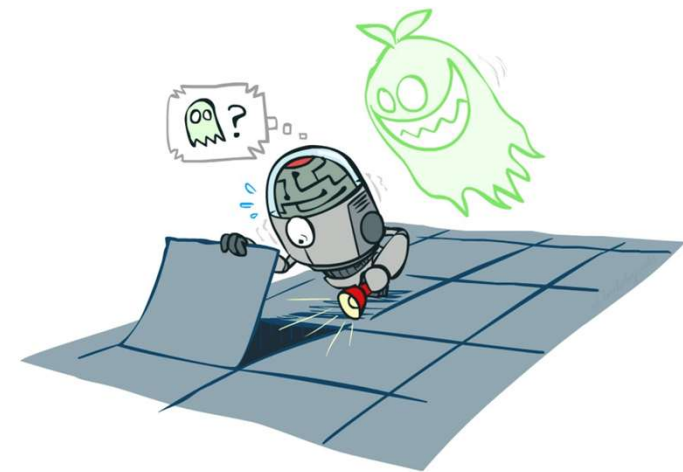
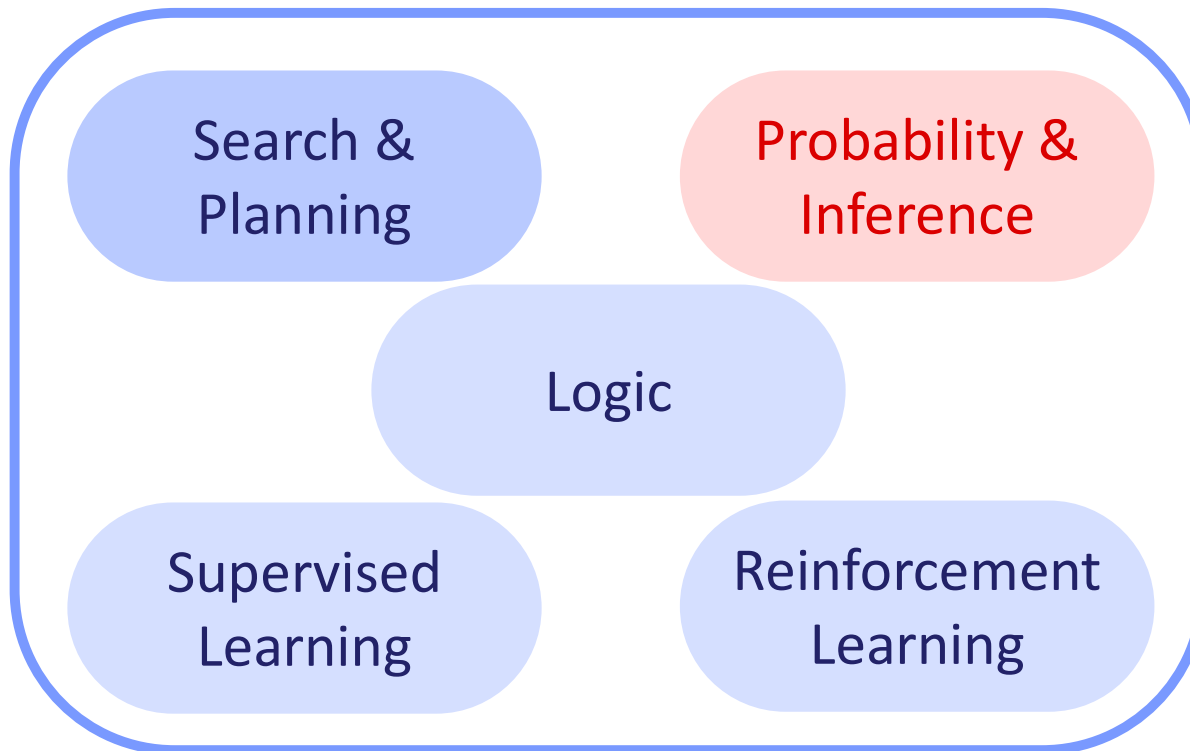
Reinforcement
Learning

Course Topics



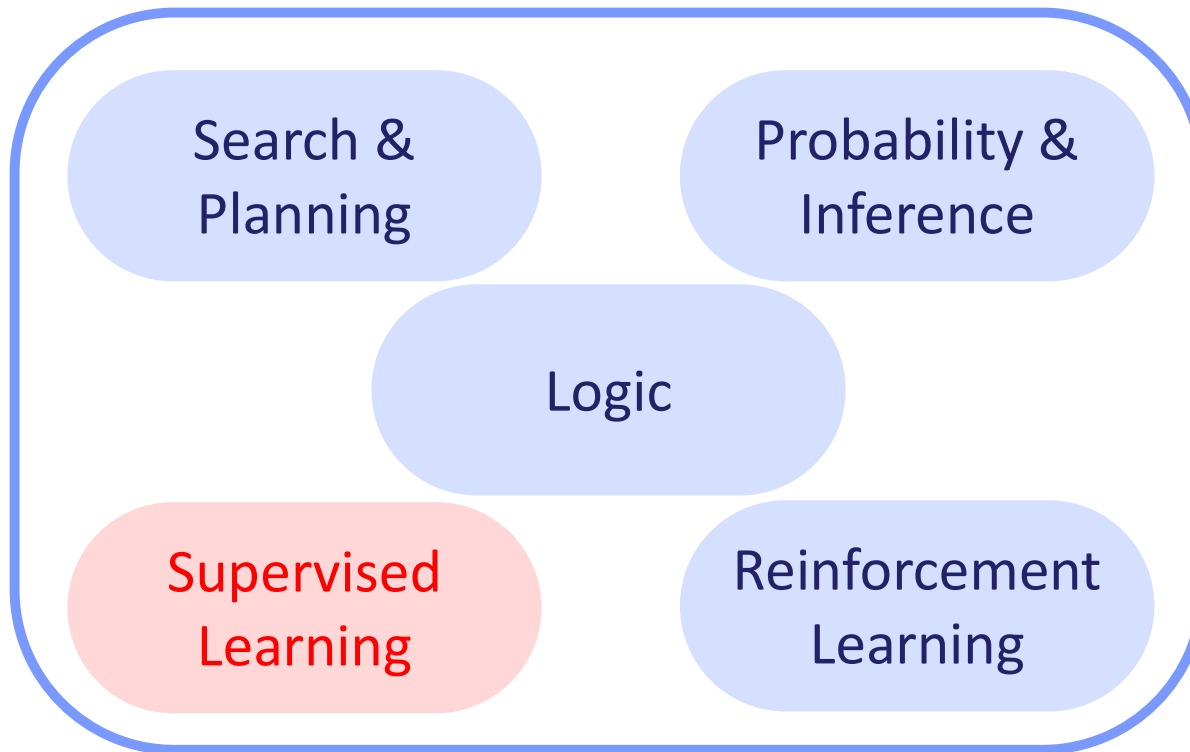
How can I use my ***model*** of the world to find a ***sequence of actions*** to achieve my ***goal***?

Course Topics



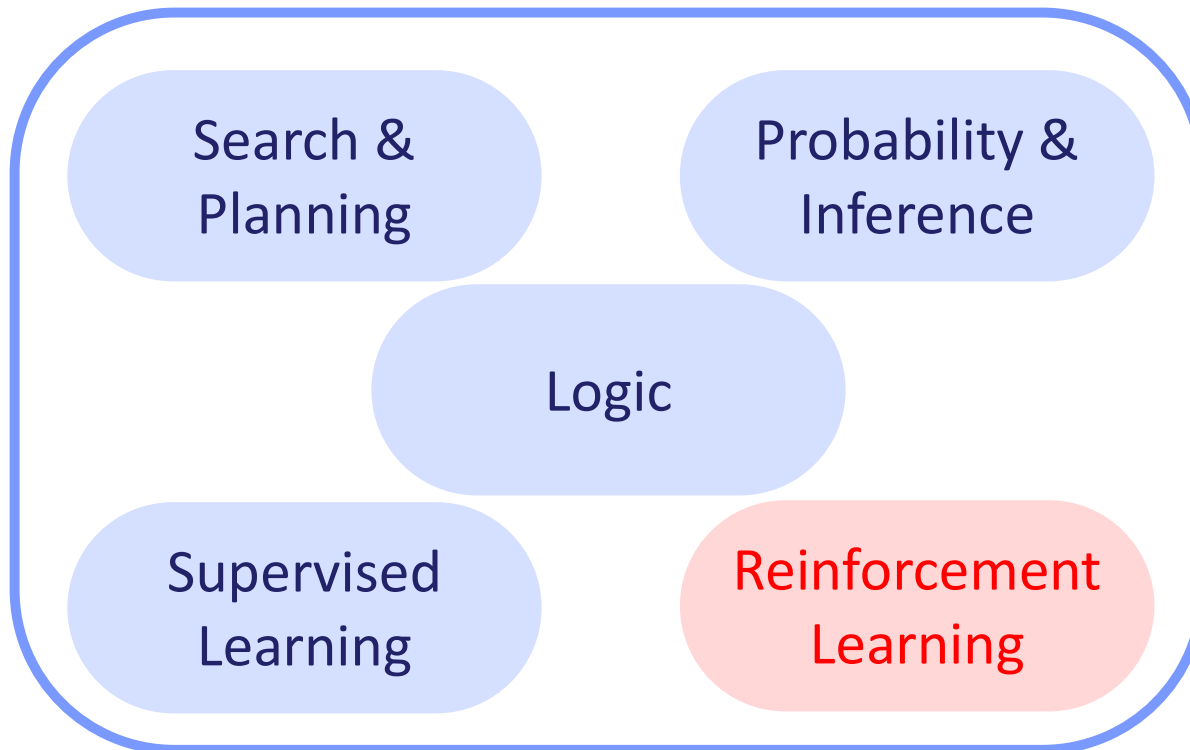
How can I make sense of *uncertainty*?

Course Topics

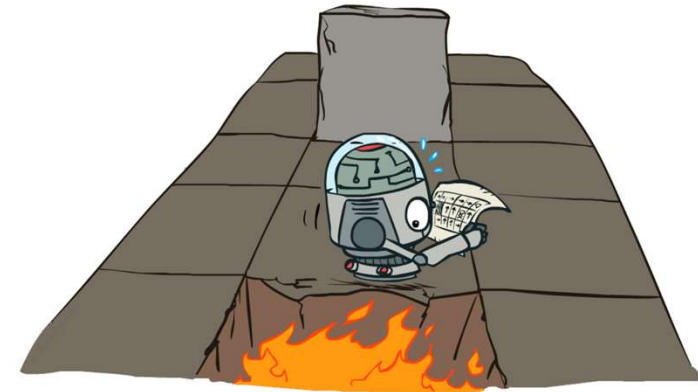


How can I learn a ***model*** of the world from ***data***?

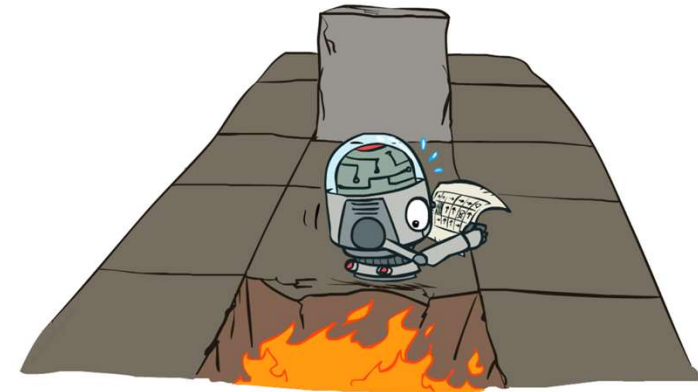
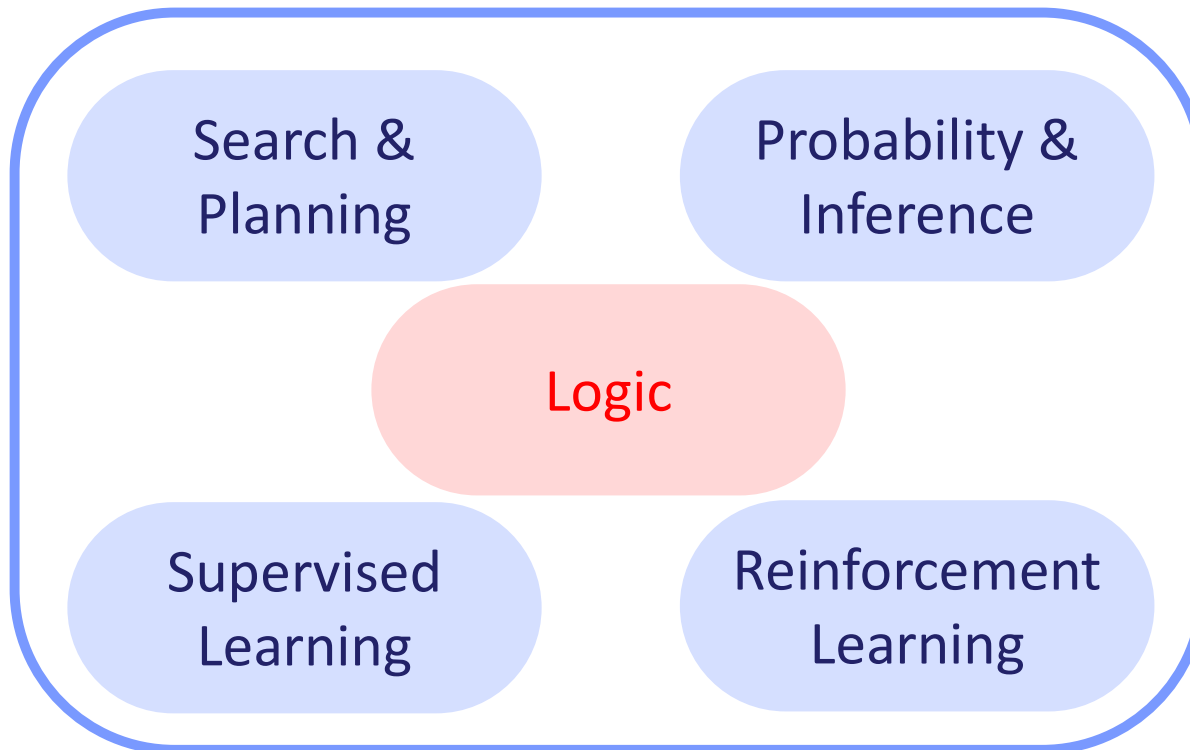
Course Topics



How can I learn a ***policy*** for any situation
so that I can ***maximize utility***?



Course Topics



How can I use **rules** and **facts** about the world to infer **new knowledge** or **make decisions**?

Next Lecture: Search

